



# Introduction: Neural Operators

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# Outline of Today

| Time  | Topics                                                             |
|-------|--------------------------------------------------------------------|
| 14:00 | Introduction & Welcome                                             |
|       | Surrogate models vs. digital twins                                 |
|       | Types & Implementations of surrogates                              |
| 14:15 | Recap of Machine Learning (ML) and Deep Learning                   |
|       | Basic concepts, evaluation, hyper-parameter tuning, etc.           |
|       | Surrogate models: Challenges for ML                                |
|       | Data-driven mitigation strategies (i.e., scenario generation)      |
| 14:45 | Physics-informed/inspired Methods -- Part I                        |
|       | Intro to Physics-informed Neural Networks (PINNs)                  |
|       | <b>Hands-On:</b> How to build and train a PINN in Python           |
| 15:30 | Coffee Break                                                       |
| 16:00 | Physics-informed/inspired Methods -- Part II                       |
|       | Operators vs. PINNs                                                |
| 16:25 | Case study: SOTA Hydraulic Surrogate Model                         |
|       | Hydraulic surrogates and their applications                        |
|       | Analyzing the problem and identifying relevant quantities and laws |
|       | A SOTA Hydraulic Surrogate                                         |
|       | <b>Hands-On:</b> Usage and Evaluation of the Surrogate in Python   |
| 17:15 | Closing, Summary, and Q&A                                          |
| 17:30 | END                                                                |



We are here!

# Extending PINNs

- PINNs Input: Space-Time Coordinate
- PDE solution only valid for a specific setup
  - Initial- / Boundary Conditions
  - PDE Parameters
- Of limited use for surrogate models
- Neural Operators address this



# What's an Operator?

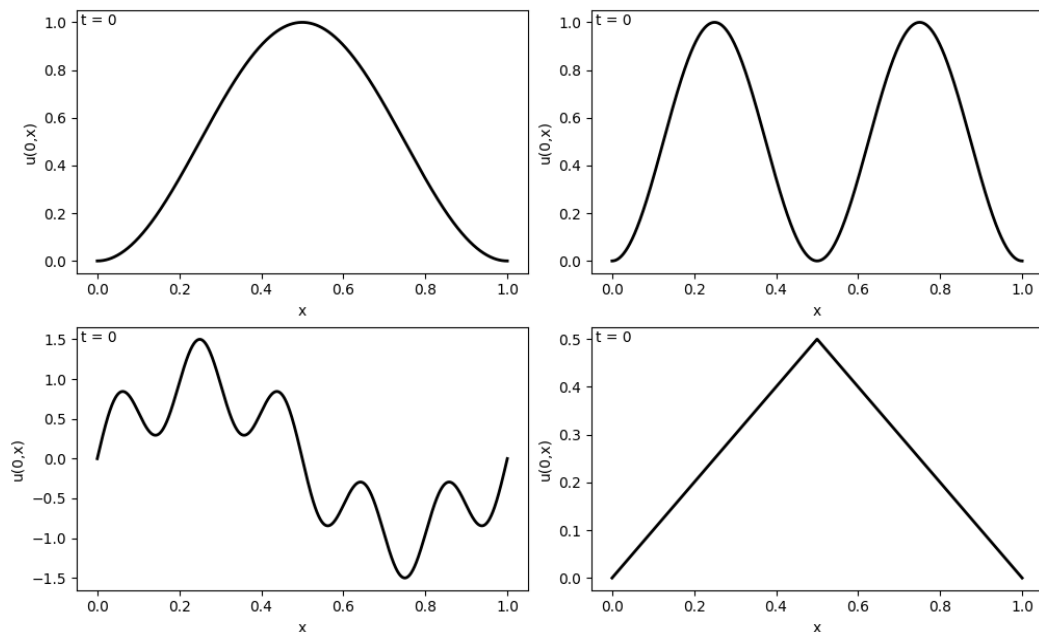
- A function acting on functions
- Example Operators:
  - Derivative
  - Laplacian
  - Fourier Transform
  - PDEs in general

$$\begin{array}{ccccc} F & : & A & \rightarrow & B \\ \uparrow & & \uparrow & & \uparrow \\ \text{Operator} & & \text{Function Spaces} & & \end{array}$$

# Solution-Operators for PDEs

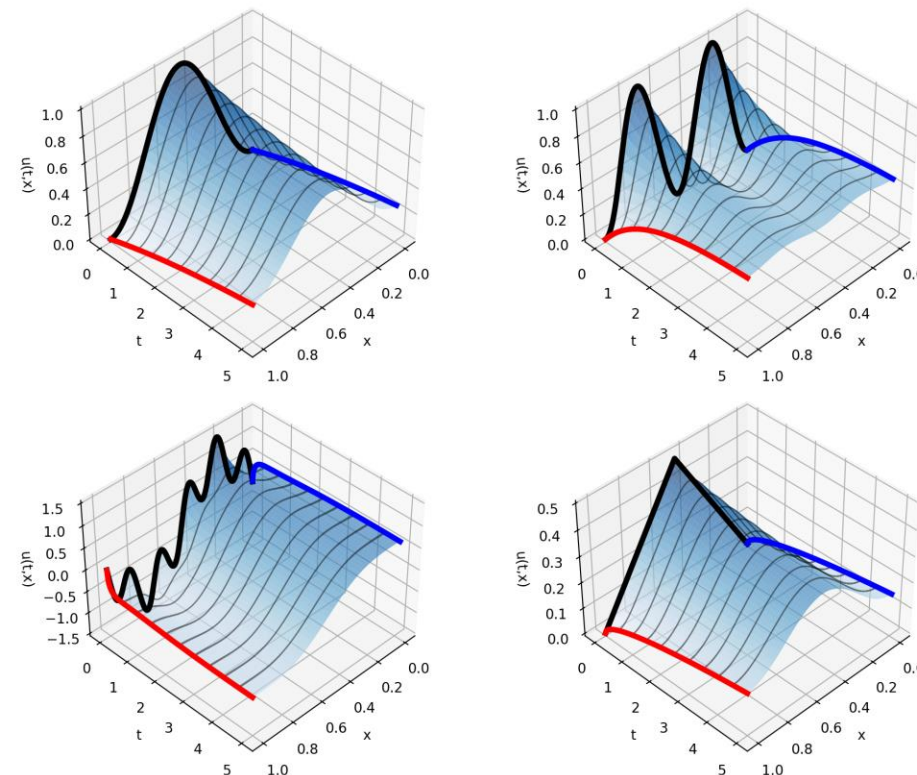
$$F : A \rightarrow B$$

**A: Space of Initial Conditions**



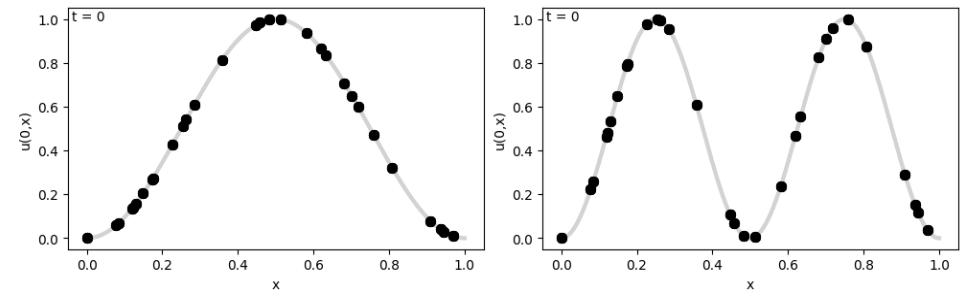
**B: Space of Solution Functions**

$F$

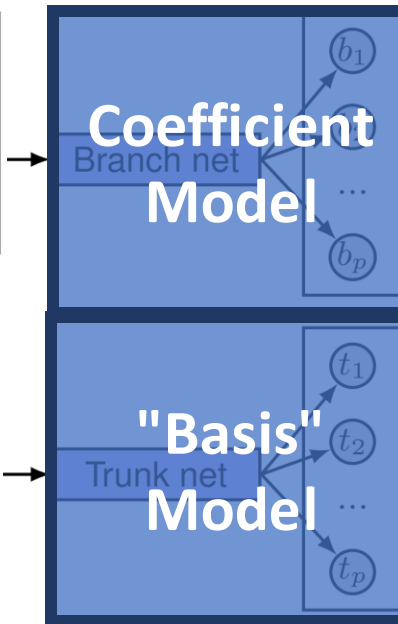
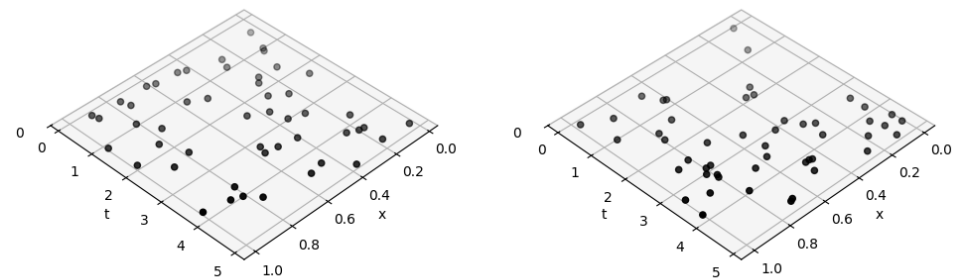


# DeepONet (Lu et al. 2019)

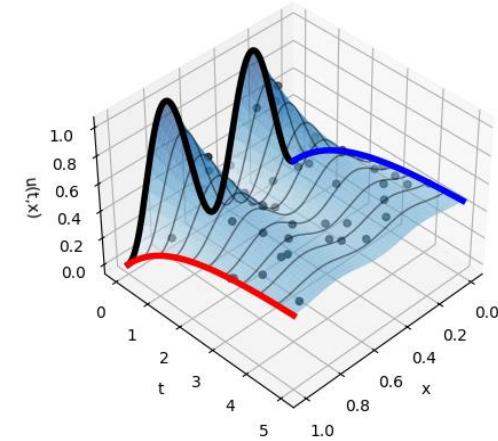
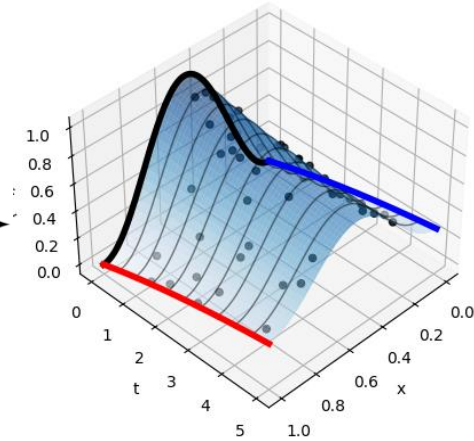
## Initial Conditions



## Space-Time Coordinates (x, t)

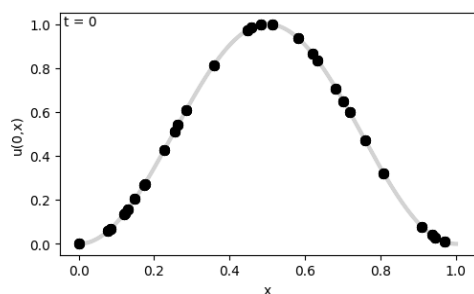


## Solution evaluated at (x, t)

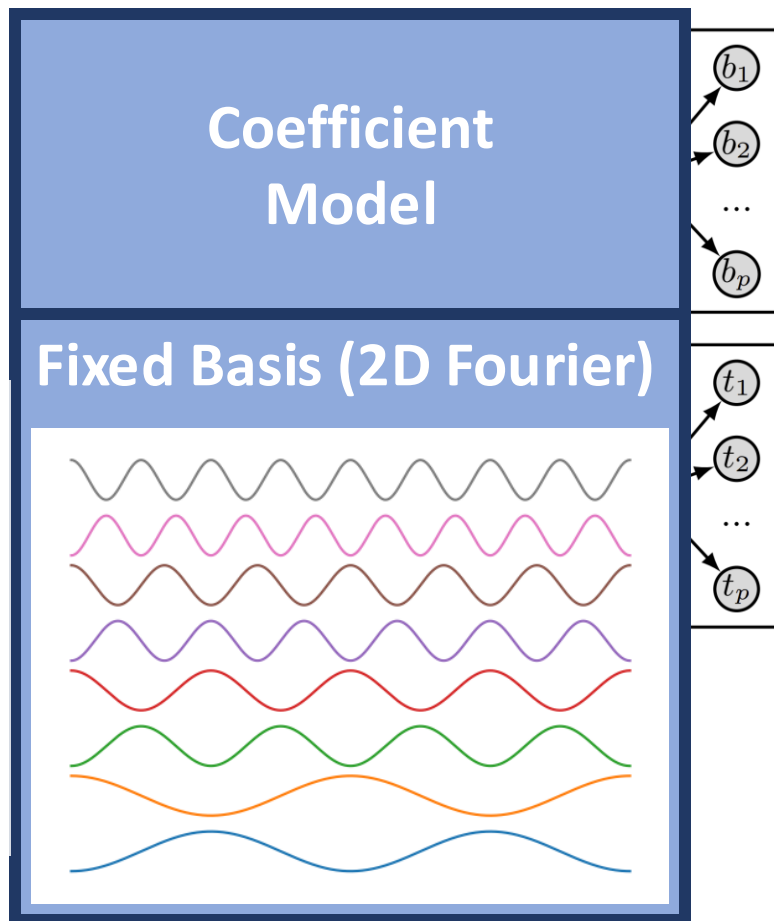
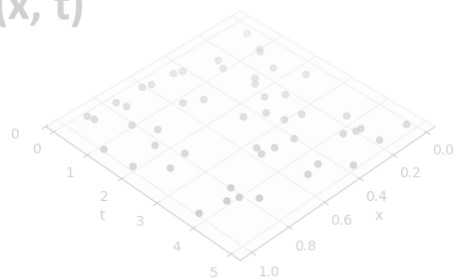


# DeepONet (Lu et al. 2019) (Simplification)

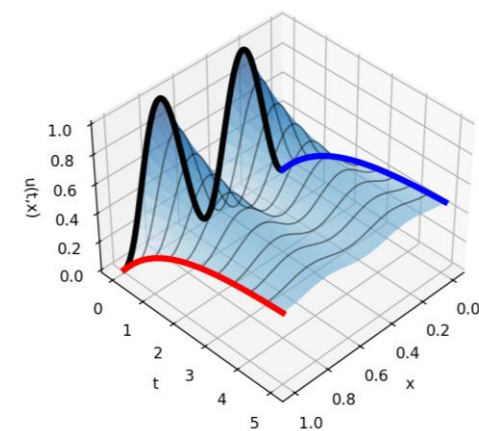
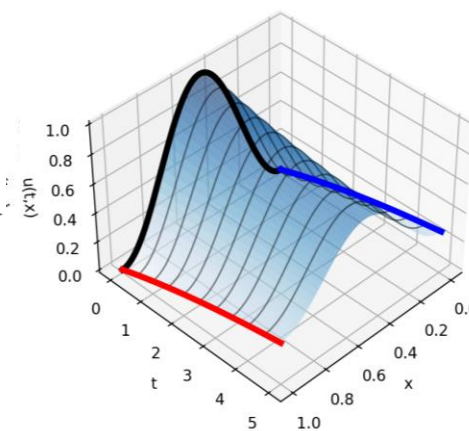
## Initial Conditions



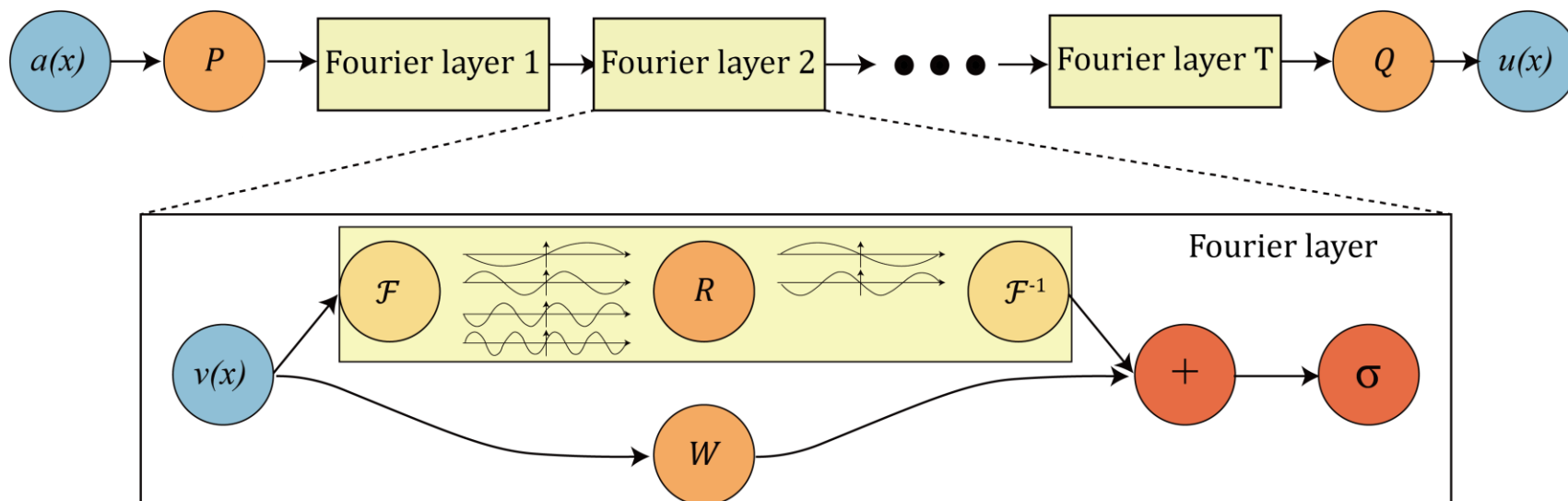
## Space-Time Coordinates $(x, t)$



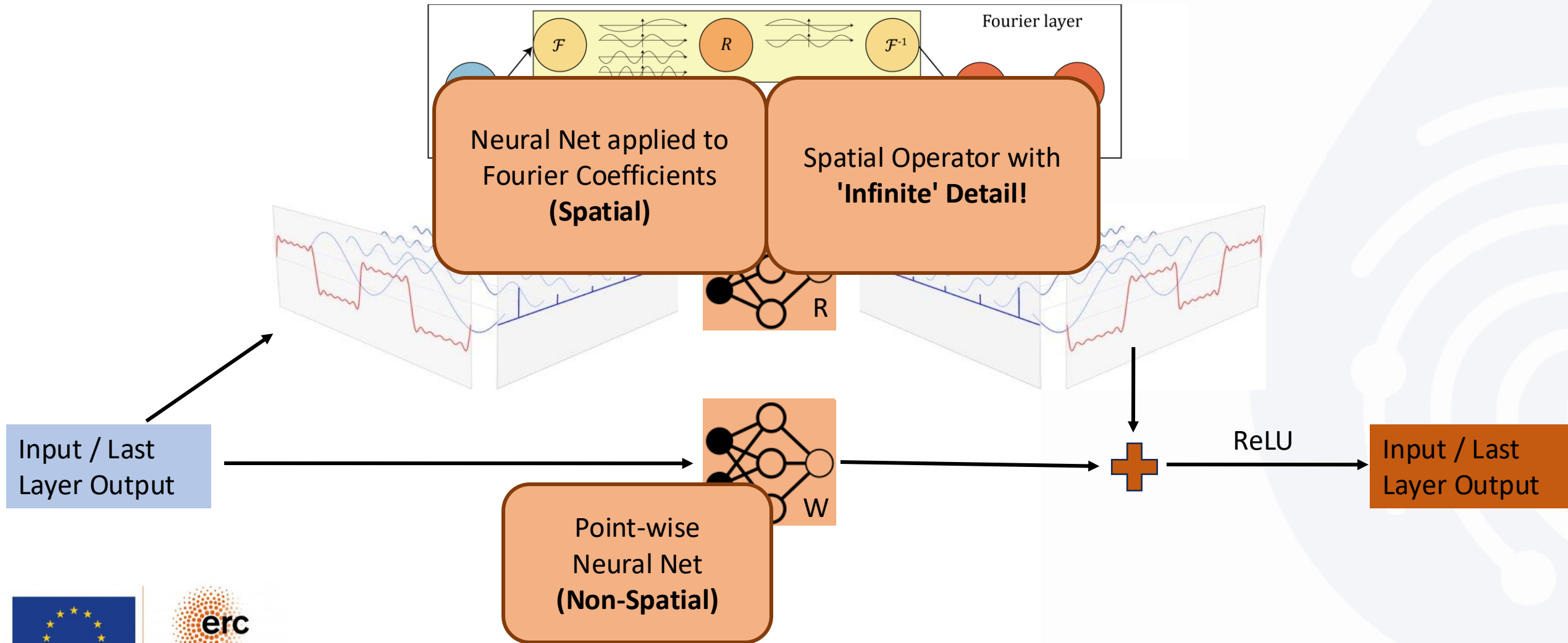
## Solution Functions



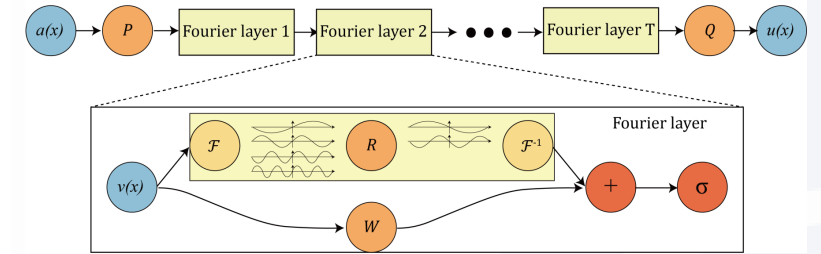
# Fourier Neural Operator (Li et al. 2021)



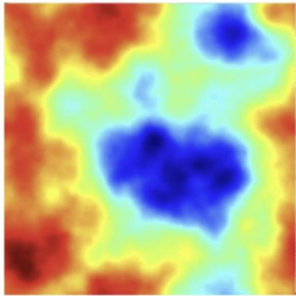
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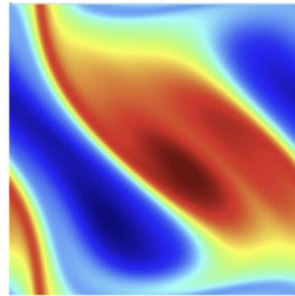
# Fourier Neural Operator (Li et al. 2021)



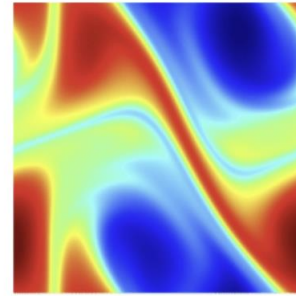
*Initial Vorticity*



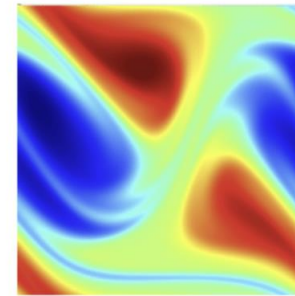
$t=15$



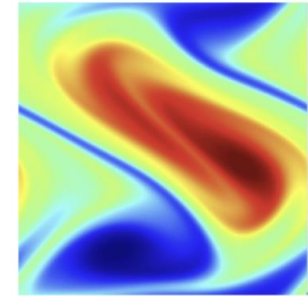
$t=20$



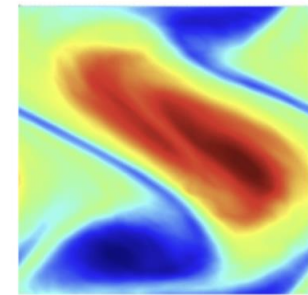
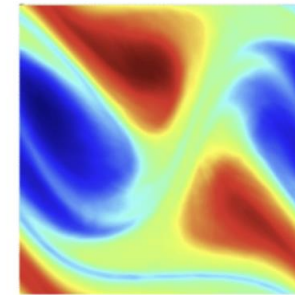
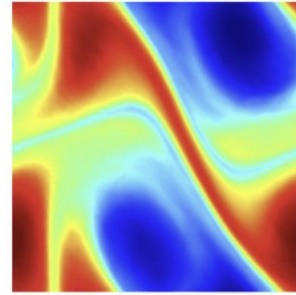
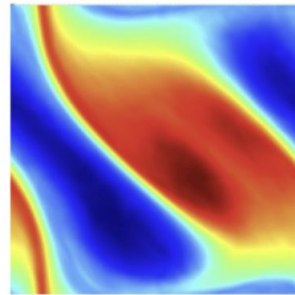
$t=25$



$t=30$



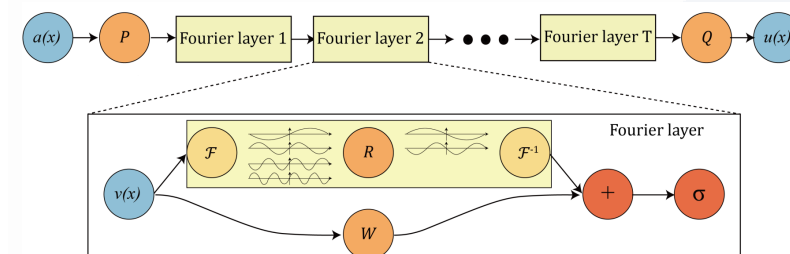
*Prediction*



Zero-shot super-resolution: Navier-Stokes Equation with viscosity  $\nu = 1e-4$ ; Ground truth on top and prediction on bottom; trained on  $64 \times 64 \times 20$  dataset; evaluated on  $256 \times 256 \times 80$  (see Section 5.4).

# Fourier Neural Operator [\(Li et al. 2021\)](#)

- Bias towards Periodic Boundaries
- Requires a regular input domain (where FFT is defined)
- Zero-Short Super-Resolution!
- Very Fast!



# Overview

## Feature

### Data Requirement

### Geometry & Mesh

### Empirical Performance

### Training Regiemes

### Practical Use Case

## FNO

High-quality simulation data

Rigid, structured grids  
(req. Of FFT)

High accuracy even for turbulent  
physics

Standard supervised loss, very  
efficient

Global fluid dynamics, weather  
forecasting, and structured grid  
problems where data can be  
generated.

## DeepONet

Explicit mathematical PDE  
equations + Data (optional)

Arbitrary, irregular, and complex  
boundaries

Less accurate / More blurry

Supervised / Semisupervised

Engineering design parts with  
highly irregular geometries,  
stress/strain models, and data-  
scarce scenarios.

# Key Take-Aways

- Universal Function Approximation extends to Operators
- Neural Operator Idea useful for Surrogates
  - Generalization to many PDE Setups (ICs, BCs, PDE Parameters)
- Typically Surrogates are Operators





# Questions?

Next: Case-Study on a Hydraulic Surrogate Model

